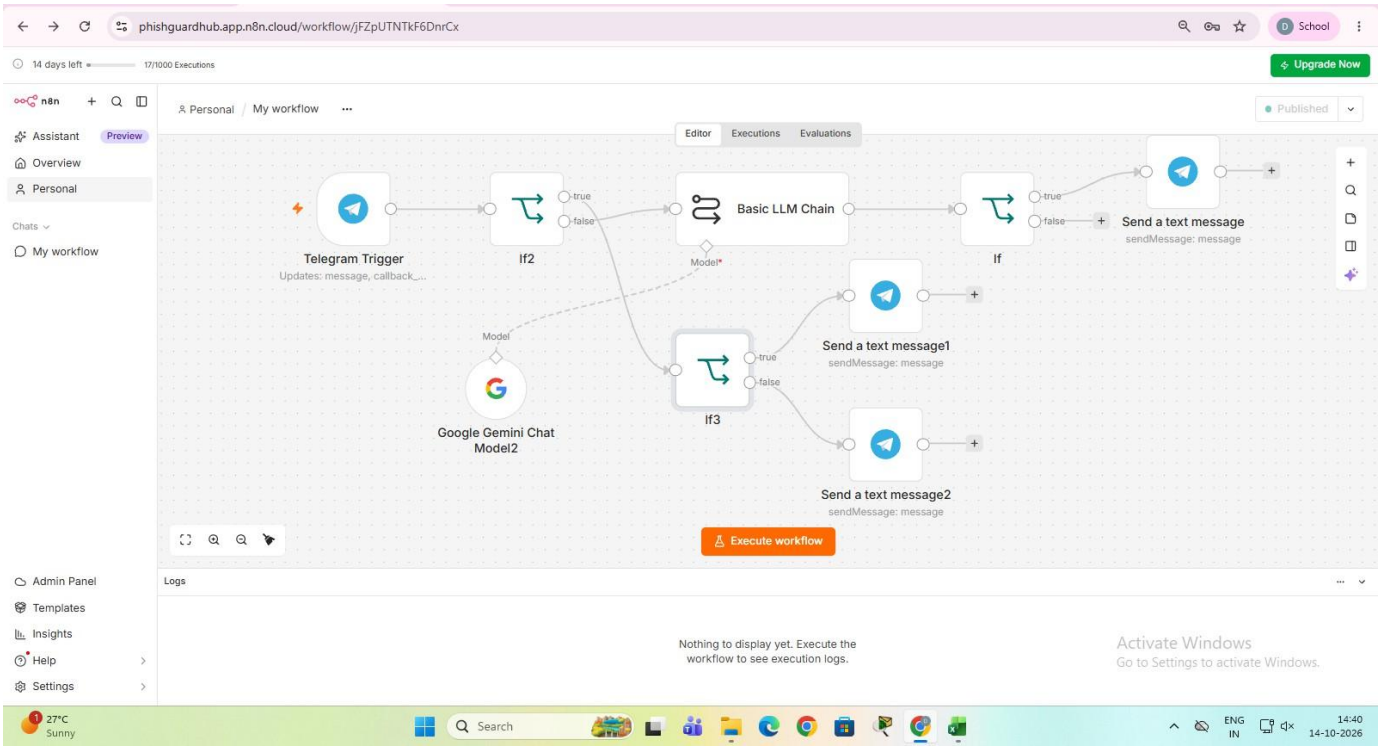
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NGP-009: MITRE ATT&CK TECHNIQUE MAPPER
Mini-Project Report

ABSTRACT

Cybersecurity analysts study suspicious activities to understand how attackers may behave inside computer systems. Mapping these activities to the MITRE ATT&CK framework can take time and requires knowledge of many techniques. This project develops an AI-assisted MITRE ATT&CK Technique Mapper using n8n workflow automation, a Telegram bot, and Google Gemini. A user sends a description or technique ID through Telegram. The workflow sends the message to the language model, which proposes a technique ID, technique name, tactic, confidence, severity, and explanation. A validation step checks the technique-ID format before the result is sent back to Telegram with APPROVE and REJECT buttons. A human reviewer can choose whether to approve or reject the suggested analysis. Five test cases were used to examine a valid ID, an invalid ID, a prompt-injection attempt, an ambiguous description, and a PowerShell example. The results show how AI and automation can organize preliminary analysis while keeping a human in the review process. The suggestions are not a substitute for checking the official MITRE ATT&CK knowledge base.

WORKFLOW



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